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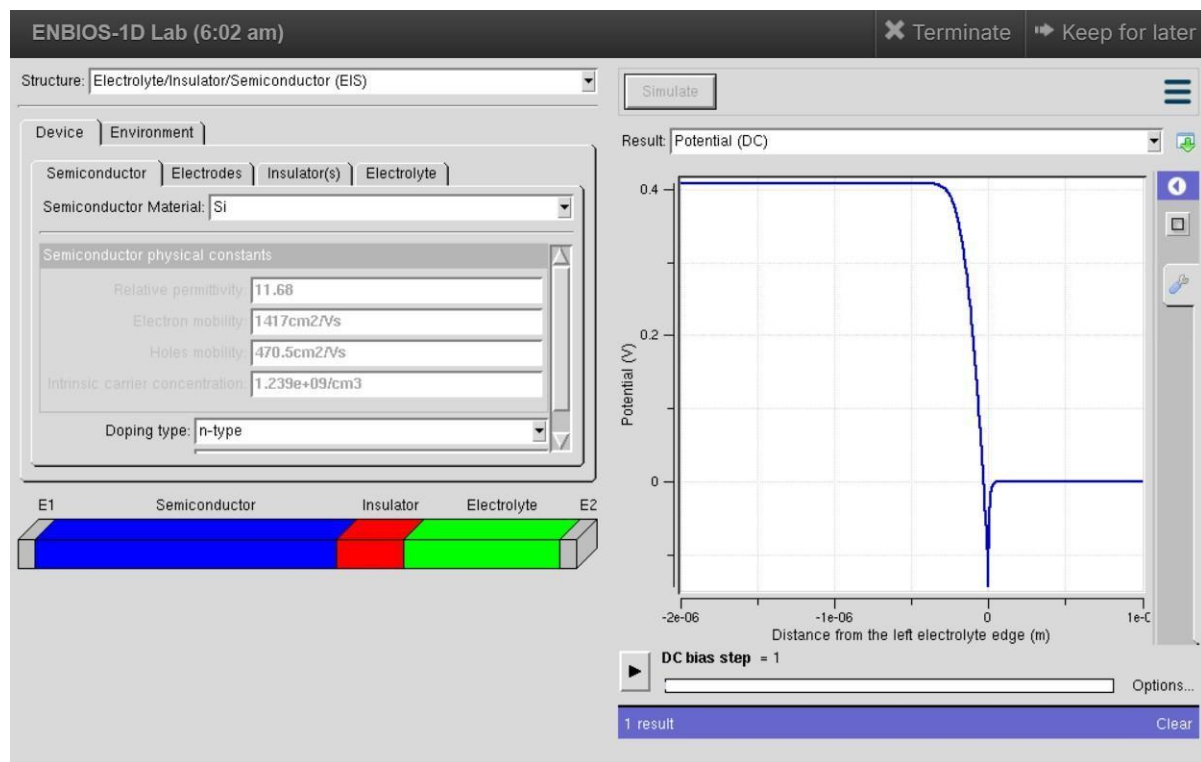
ECA3102

Nanoelectronics

Exp 10: Calculate the Potential of a pH Sensor using NanoHub

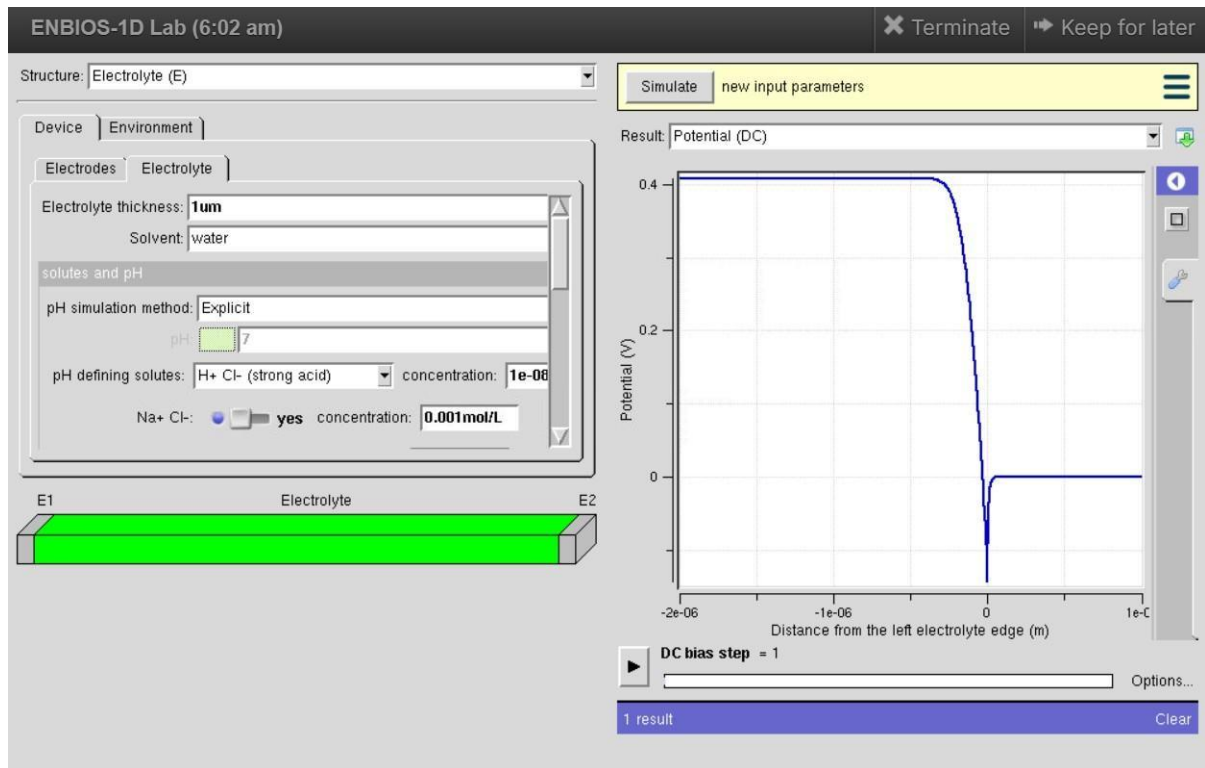
Aim: To obtain the potential of a sensor by varying the pH value of an electrolyte and insulator Thickness.

Case 1: Selected Semiconductor/insulator/electrolyte as structure



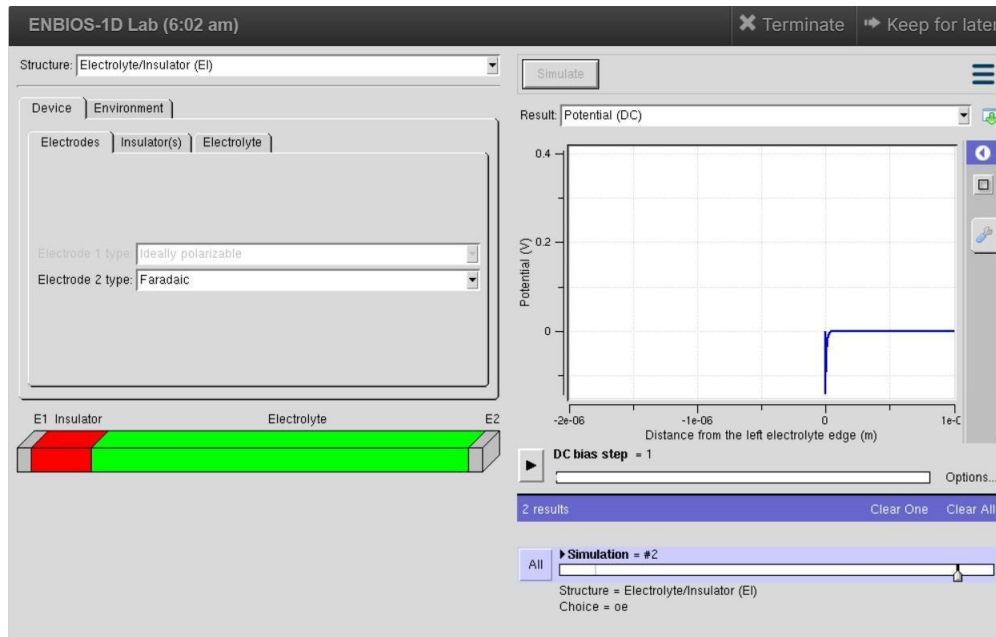
The simulated potential remains nearly constant at approximately 0.4 V throughout the semiconductor region, followed by a sharp potential drop at the insulator–electrolyte interface, after which the potential becomes close to 0 V in the electrolyte. This narrow transition indicates a strong electric field across the insulating layer. In an EIS structure, the insulator electrically isolates the semiconductor from the electrolyte while allowing electrostatic coupling. The electrolyte pH changes the surface charge at the insulator interface, which shifts the surface potential and causes changes in the surface band bending of the n-type semiconductor. The sharp potential variation confirms efficient electrostatic coupling and shows that the EIS structure provides high sensitivity and accuracy for pH sensing.

Case 2: Selected Electrolyte as Structure



The electrolyte initially maintains a nearly constant potential of about 0.4 V, while the potential decreases rapidly to approximately 0 V near the electrode boundary and remains almost uniform throughout the rest of the electrolyte. Since this structure contains only the electrolyte, the potential mainly depends on ionic distribution and electrode potential. At pH 7, the stable ionic concentration produces an almost uniform potential, while the sharp drop near the electrode is due to charge accumulation forming an electrical double layer at the electrolyte–electrode interface. Since there is no semiconductor or insulator, there is no mechanism for surface-potential amplification or carrier modulation. Therefore, the electrolyte-only structure has limited pH sensing capability and mainly represents electrochemical equilibrium rather than semiconductor-based sensing.

Case 3: Selected Electrolyte/Insulator as Structure



In the EI structure, the potential remains close to 0 V over most of the structure, with only a small negative potential dip at the electrolyte–insulator interface. The potential variation is much smaller than that observed in the EIS structure because the semiconductor is absent. The insulator blocks direct electrical conduction, and without a semiconductor there is no carrier modulation or band bending. The small potential variation is mainly caused by interface charge accumulation and the electric field across the dielectric layer. Increasing the insulator thickness would further reduce electric-field penetration and decrease sensitivity. Hence, the EI structure alone cannot provide effective pH sensing because it lacks the semiconductor-based electrical amplification required for efficient sensing.

Comparison

The three structures show clear differences in their potential response and sensing capability. The EIS structure produces the largest and sharpest potential variation because the semiconductor enables carrier modulation and converts pH-induced surface-potential changes into measurable electrical signals. The electrolyte-only structure shows an almost uniform potential with a boundary drop caused mainly by ionic equilibrium and the electrical double layer, resulting in limited sensing capability. The EI structure produces only a small interface potential because it has an insulator but no semiconductor for carrier modulation and electrical amplification. Thus, the semiconductor, insulator, and electrolyte each play an important role: the electrolyte provides the pH-dependent surface charge, the insulator provides electrical isolation and electric-field control, and the semiconductor converts the surface-potential change into a measurable electrical response.

Conclusion

The NanoHub simulation successfully demonstrates the working principle of a pH sensor using different structural configurations. Among the three cases, the Electrolyte–Insulator–Semiconductor (EIS) structure provides the highest potential sensitivity and the most pronounced interface potential variation, making it the most suitable configuration for pH sensing. The electrolyte-only and EI structures show lower sensitivity because they lack semiconductor-based carrier modulation. Therefore, an optimized EIS structure with suitable insulator thickness and controlled electrolyte pH provides the best performance for highly sensitive pH sensor and BioFET applications.